higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T380**(E)**(M28)T

# NATIONAL CERTIFICATE

# COMMUNICATION-ELECTRONICS N4

(8080224)

**28 March 2019 (X-Paper)**
**09:00–12:00**

**Calculators may be used.**

**This question paper consists of 6 pages and 1 formula sheet.**

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# DEPARTMENT OF HIGHER EDUCATION AND TRAINING
# REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
COMMUNICATION-ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

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**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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## QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 An antenna in any receiver converts audio to sound.

1.2 Ultraviolet radiation from the sun is the primary cause of ionisation.

1.3 A microphone converts audio signals into electromagnetic waves.

1.4 The function of the catcher cavity in a klystron amplifier is to emit a stream of electrons.

1.5 The impedance of a series circuit at resonance is a minimum.

1.6 An AM signal with 110% modulation is over-modulated.

1.7 A multicavity magnetron is an example of a modulator in radio communication.

1.8 Construction of a rhombic antenna is very cheap.

1.9 Radio waves travel at a speed of $3 \times 10^6$ metres per second.

1.10 A radio antenna is directional.

(10 × 1) **[10]**

## QUESTION 2

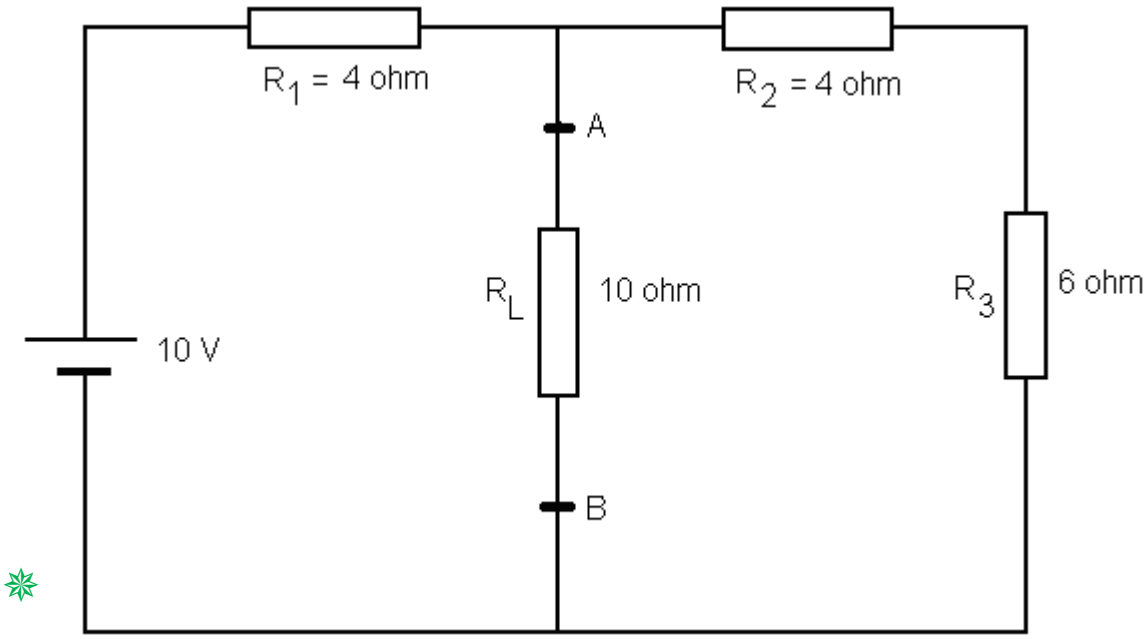
2.1 Define each of the following:

2.1.1 Kirchhoff's current law

2.1.2 Thévenin's theorem

(2 × 2) (4)

2.2 Use Thévenin's theorem to find the equivalent circuit of the network shown below. Draw the equivalent network circuit.




(6)
**[10]**

**QUESTION 3**

A parallel circuit consists of a coil with an internal DC resistance of 10 Ω and an inductance of 15 mH in series with a 200 nF capacitor.

3.1 Calculate the:

3.1.1 Total impedance

3.1.2 Resonance frequency

3.1.3 Bandwidth

3.1.4 Q-factor

(4 × 2) (8)

3.2 Prove the answer in QUESTION 3.1.4 by using any other alternative method. (2)
**[10]**

## QUESTION 4

4.1 Make neat sketches of each of the following attenuators:

4.1.1 O-type

4.1.2 H-type

(2 × 2) (4)

4.2 Draw a circuit diagram of a band-pass filter. (3)

4.3 Draw the basic attenuation versus frequency graph for the filter in QUESTION 4.2 (3)

**[10]**

## QUESTION 5

Make neat, fully labelled sketches of each of the following:

5.1 Travelling wave tube

5.2 Reflex oscillator

(5 × 2) **[10]**

## QUESTION 6

6.1 Draw a fully labelled block diagram of an AM-transmitter. (5)

6.2 Briefly explain the function of each stage of the transmitter in QUESTION 6.1. (10)

6.3 Draw a circuit diagram of an AM full-wave detector. (5)

**[20]**

## QUESTION 7

7.1 Draw a block diagram of an FM-superheterodyne receiver. (5)

7.2 Make a neat, labelled circuit diagram of an FM-modulator. (5)

7.3 Define *frequency modulation*. (2)

7.4 Define *modulation index*. (2)

7.5 What is the function of the circuit diagram in QUESTION 7.2? (1)

**[15]**

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## QUESTION 8

8.1 Determine the length of a standard half-wave dipole antenna used to receive a 60 MHz radio signal. Assume that the velocity of the electromagnetic wave in the antenna is $3 \times 10^8$ m/s. (2)

8.2 What is meant by *vertical polarisation* with regard to antennae? (1)

8.3 Explain the term *ionosphere* as applicable to radio-wave propagation. (2)

8.4 Show by means of a sketch the possible ionosphere layers during the day as well as at night. (4)

8.5 Define *critical angle* as applicable to radio-wave propagation. (2)

8.6 Make neat, labelled sketches of each of the following antennae:

8.6.1 Loop antenna

8.6.2 Standard half-wave dipole antenna

(2 × 2) (4)

**[15]**

**TOTAL: 100**

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## FORMULA SHEET

1. $I = \frac{V}{R}$
2. $X_L = 2\pi fL$
3. $X_C = \frac{1}{2\pi fc}$
4. $Z = \sqrt{R^2 + (X_L - X_C)^2}$
5. $I = \frac{V}{Z}$
6. $V_R = I \,.\, R$

   $V_L = I \,.\, X_L$

   $V_C = I \,.\, X_C$
7. $Cos\phi = \frac{R}{Z}$
8. $I = \sqrt{I_R + I_C^2}$
9. $\theta = Tan^{-1}\frac{I_C}{I_R}$
10. $I = \sqrt{I_R^2 + I_L^2}$
11. $I = \sqrt{I_R^2 \ + (I_L - L_C)^2}$
12. $\theta = Tan^{-1}\frac{(I_L - I_C)}{I_R}$
13. $f_r = \frac{1}{2\pi\sqrt{LC}}$
14. $C = \frac{1}{4 \,.\, \pi^2 \,.\, f_r^2 \,.\, L}$
15. $L = \frac{1}{4 \,.\, \pi^2 \,.\, f_r^2 \,.\, C}$
16. $Q = \frac{X_L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_R}{bandwidth/bandwydte}$
17. $V_c = V \times Q \; and/en \; V_L = V \times Q$
18. $Z = \frac{L}{CR}$
19. $I_c = I \times Q$
20. $I_L = I \times Q$
21. $f = \frac{\nu}{\lambda}$
22. $B - W = \frac{R}{2\pi L}$
23. $V_T = \sqrt{V_R{}^2 + (V_C - V_L)^2}$

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